

Chandramita Bhattacharya

AI Engineer | Agentic Applications & Enterprise GenAI

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Profile

AI Engineer with an M.Sc. in Cognitive Systems and over four years of experience across enterprise software engineering, full-stack development, and applied AI. Delivered business-critical enterprise applications at PwC and currently developing agentic AI workflows at Liebherr for knowledge retrieval, document processing, and operational automation. Working in end-to-end delivery, from requirements analysis and solution design to implementation, testing, evaluation, and cloud deployment, with a focus on RAG, LLM integration, automated testing, and stakeholder collaboration.

Education

M.Sc. Cognitive Systems, Ulm University, Germany | 2022 - 2025 | Grade: 1.5 | Thesis: 1.0

B.Tech Computer Science, Institute of Engineering and Management, India | 2015 - 2019 | Grade 8.7/10

Technical Skills

AI & ML Engineering: Python, LangChain, LangGraph, Azure OpenAI, RAG, MCP, PyTorch, scikit-learn

Retrieval, Evaluation & LLMops/MLOps: Azure AI Search, Mosaic AI Vector Search, Databricks, Langfuse, LiteLLM, Ragas

Full-Stack & Data: FastAPI, Pydantic, Angular, React, TypeScript, JavaScript, Node.js, REST APIs, SQL, PostgreSQL, pgvector

Cloud, DevOps & Testing: Azure, Docker, Kubernetes, Git, GitHub, Jira, CI/CD, pytest, pytest-asyncio, Agile

Experience

Data & AI Engineer Intern | Liebherr Digital Center, Ulm, Germany

07/2025 - Present

- Developed an agentic knowledge assistant using LangChain components, LangGraph for orchestration, LiteLLM for model routing and Langfuse for tracing, reducing manual search effort and enabling faster knowledge access.
- Compared custom RAG pipelines with Databricks AI Functions and curated a specialised table-parsing benchmark dataset, identifying preprocessing and retrieval configurations that improved overall RAG accuracy.
- Implemented a document-intelligence pipeline using YOLO, SAM, and PaddleOCR to extract complex layouts and table structures from scanned technical documents, reducing reliance on time-consuming manual extraction.
- Developed PyTorch-based YOLOX pedestrian detection and fish-eye video frame undistortion to improve construction site safety and reduce worker exposure to hazards for the LiXplore project.

Associate, Technology Consulting | PricewaterhouseCoopers (PwC), India

07/2019 - 08/2022

- Delivered production enterprise applications using Angular, TypeScript and JavaScript within Agile teams, translating business requirements into tested and maintainable software releases.
- Modernised a legacy reconciliation workflow into a secure, role-based platform with validation, reporting, and automated processing across multiple data sources, improving operational control and workflow consistency.
- Implemented Google Analytics and Google Tag Manager tracking for an automotive e-commerce platform, enabling user-behaviour analysis and data-informed product improvements.

Selected Projects

Master's Thesis | Visual Computing Group, Ulm University

12/2024 - 08/2025

- Benchmarked six text-to-3D models across shape, colour and style using four complementary semantic, visual-plausibility and mesh-quality metrics, showing that model behaviour varied substantially based on the evaluation criterion.

Controlled Email-to-ERP Workflow | Python, Pydantic, LLMs, REST APIs, JSON

- Developed an auditable workflow that extracts order data from emails and PDF/Excel/PowerPoint attachments, validates it against ERP master data, and routes clean cases to draft creation while escalating ambiguous or high-risk requests.

Human Evaluation of AI-Generated Visual Content | Python, Gradio, pandas

- Co-designed a taxonomy and annotation protocol for 260 AI-generated portraits, reaching category consensus for 77.5% of evaluated images and showing that agreement varied across different types of visual defects.

Explainable AI Benchmarking | OpenCV, YOLO, SAM, LIME, PyTorch

- Compared five segmentation strategies for LIME explanations of an InceptionV3 convolutional neural network (CNN) across 100 images, analysing explanation quality, runtime, and segmentation behaviour.

Additional Information

Additional Experience: Research Assistant, Ulm University - multimodal material-perception and 3D-navigation research; Philips Carbon Black Ltd., India (Intern) - SAP Fiori workflow and usability analysis; Linde Global IT, India (Intern) - UiPath automation of SAP ERP maintenance.

Languages: English (C1), German (B1, in progress), Bengali (native), Hindi (fluent)